

Pandemic resilience theme
AI for Collective Intelligence
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10th Jan 2026

PLOS Computational Biology
PLOS
Nine Hills Road
Cambridge, CB2 1GE
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Dear Editor,

Re: Methods paper: Instantaneous Reproduction Number Estimation From Modelled Incidence

Please consider our submission of this paper for publication in PLOS Computational Biology.

In this paper we describe a new mathematical method for estimating the time varying reproduction number of an infectious disease outbreak, given an estimate of the generation time. Rather than using raw case count data, the method takes modelled estimates of disease incidence as input and propagates uncertainty in both modelled incidence and generation time. By decoupling the reproduction number estimation from the raw count data we address several key limitations of existing approaches. Firstly we can estimate reproduction numbers from incidence models derived from incomplete data, and are not constrained to data provided as daily timeseries. Secondly we can estimate reproduction numbers from incidence models that incorporate adjustment for any number of factors, such as weekly periodicity, or right truncation of the data due to observation delays. Incidence models are simple to construct from raw count data in a number of statistical frameworks and easy to validate and debug, which we find increases reliability of our reproduction number estimates.

In the manuscript we describe the mathematical formulation of the estimation method, and characterise its performance against simulated data from a branching process model. We compare it to the de-facto standard method for estimating the reproduction number, EpiEstim, using probabilistic forecast scores looking at bias, accuracy, forecast and classification skill.

The paper is accompanied by an R package which contains a reference implementation of the method and additional examples of its application against both real life and simulated datasets. The method is deterministic, requires no parameterisation beyond the incidence model and generation time estimates, and the implementation is highly performant on modest hardware. The package has been published on r-universe and we are preparing it for submission to CRAN.

While there are numerous methods for estimating the reproduction number, for the reasons articulated above we believe that there are significant benefits to our approach and that it can reliably inform both the response to a future pandemic and the monitoring of seasonal disease outbreaks, particularly in settings where data quality and availability of compute are limited.

We believe this manuscript and the associated R package is of interest to your readership and will be grateful if you could consider this submission for publication in PLOS Computational Biology.

Yours sincerely,

Dr. Robert Challen Ph.D. MBBS